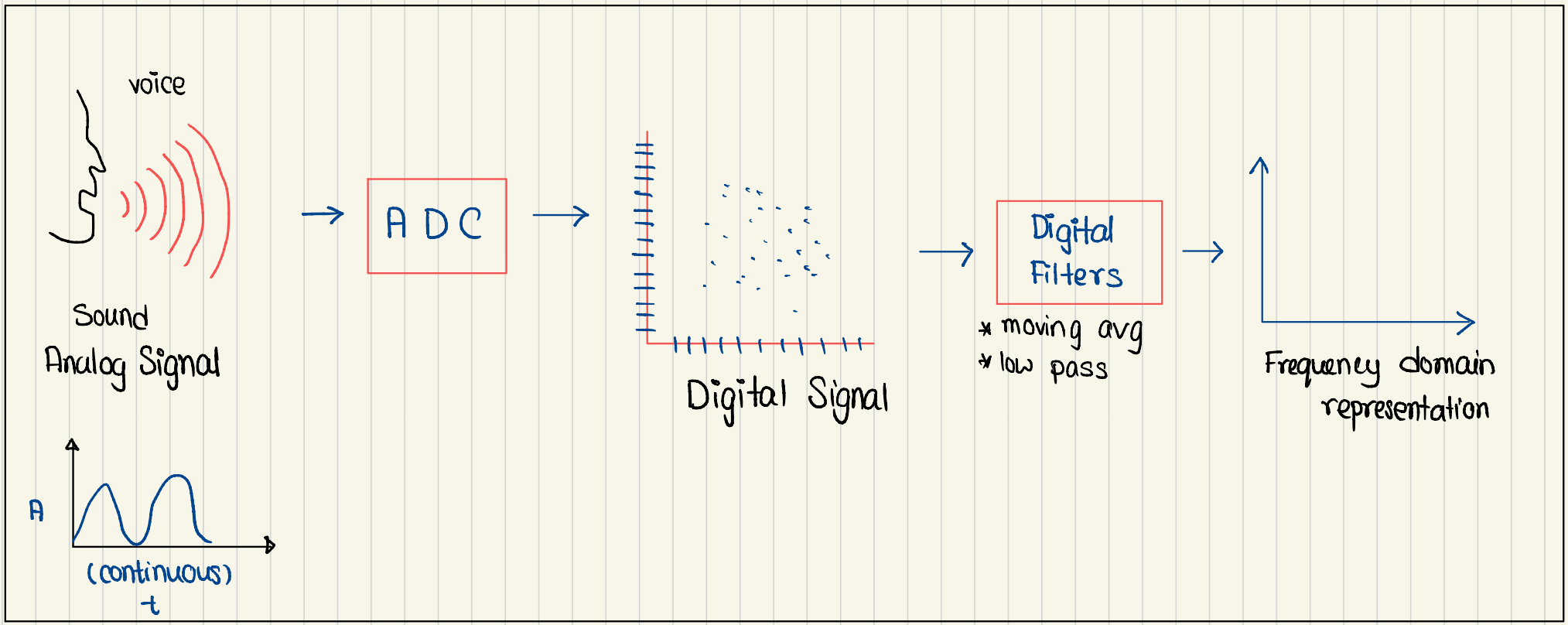
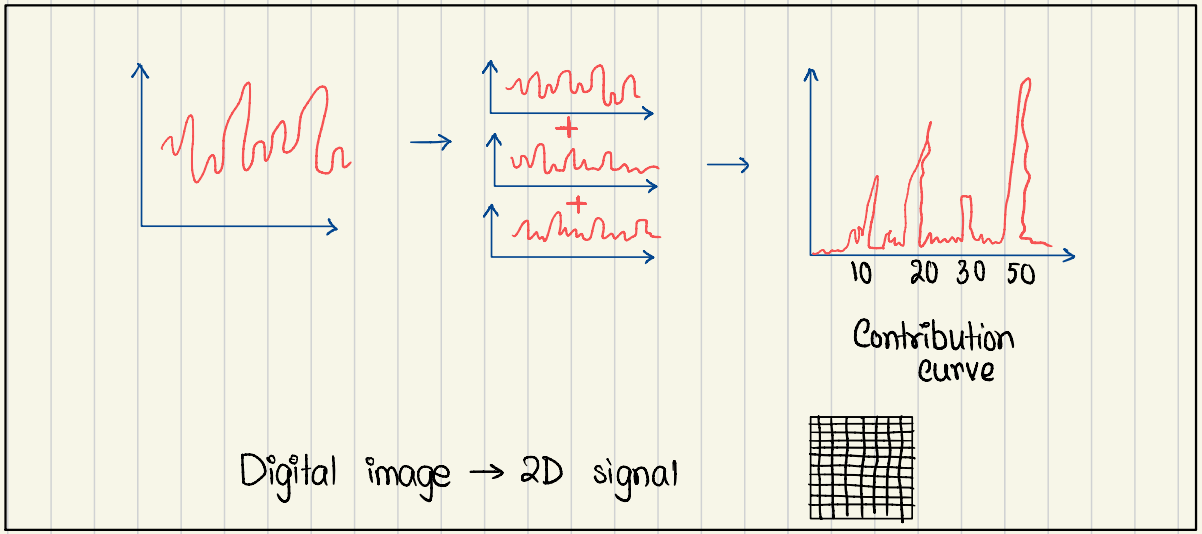
# DSP Notes

PROAKIS, CH-1

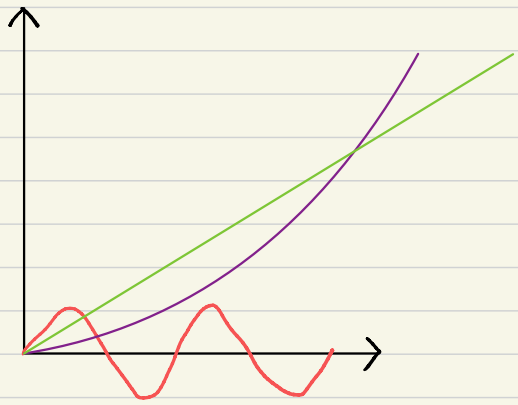

voice
Sound
Analog Signal
A
(continuous)
t
ADC
Digital Signal
Digital Filters
* moving avg
* low pass
Frequency domain representation


10 20 30 50
Contribution curve
Digital image → 2D signal

## Signal

A physical quantity that varies with time.

Function of one or more independent variables.



linear $S_1(t) = 5t$

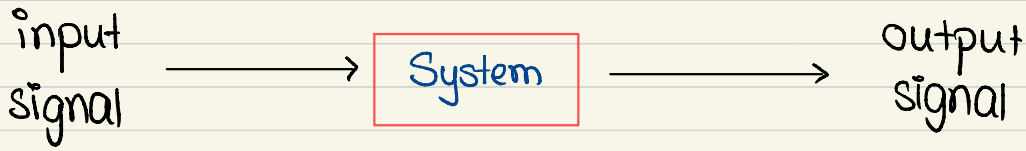
quadratic $S_2(t) = 10t^2$

trig. $S_3(t) = \sin(t)$

compound $S_4(t) = A\sin(2t) + B\cos(3t)$

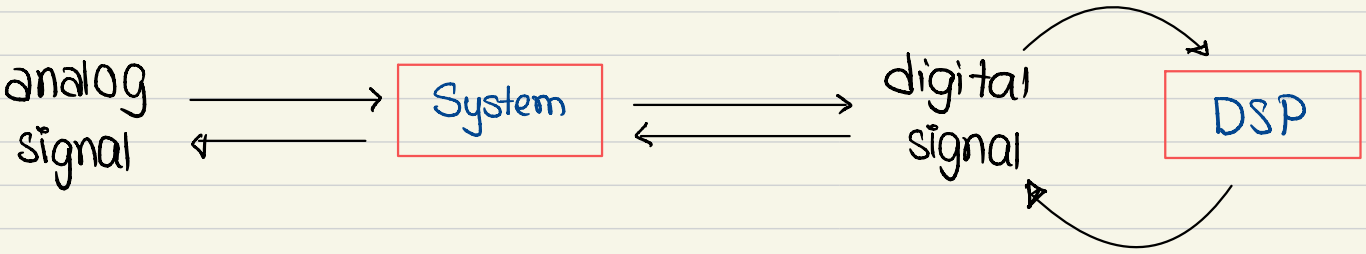
## System

A physical device that performs an operation on a signal.




Example : Digital Filter (reduce noise & interference)

ADC




Digital signal → Easier to process, store, analyze

## Basics

i) Most of the systems that we encounter are analog.

ii) Analog systems (filters, frequency analyzers/multipliers).

iii) Digital vs Analog

* Digital programmable system allow flexibility in reconfiguring the digital processing operation by changing program.
  Analog system requires redesigning the hardware.

* Higher accuracy in digital systems.

* Digital signals can be easily stored without loss of signal fidelity, making it transportable & remotely processable.

* DSP allows implementation of more sophisticated signal processing algorithms.

* DSP is cheaper.

* Limitation: Speed of operation, signals with wide bandwidth requires fast sampling rate.

$I(x,y)$ → BW Image
$I(x,y,t)$ → BW Video

## Classification

### 1. Multi-channel signal

Signals generated from multiple sensors/sources, or multiple signals generated from the same source.

$$S_3(t) = \begin{bmatrix} s_1(t) \\ s_2(t) \\ s_3(t) \end{bmatrix}$$

1 signal → multiple channel

ex - accelerometer, ECG

audio, color img, ECG

### 2. Multi-dimensional signal

Function of multiple independent variables arising from a single source

$I(x,y)$ → BW image, 2D

$I(x,y,t)$ → BW video, TV

### 3. Multi-channel Multi-dimensional Signal

Signal originating from multiple sources having multiple independent variables.

$$I(x,y,t) = \begin{bmatrix} I_r(x,y,t) \\ I_g(x,y,t) \\ I_b(x,y,t) \end{bmatrix}$$

→ Colour Video

→ Light up different pixels.

Value → output
Time → input

n

## Continuous Time Signal (Analog Signals)

Defined for every value of time

Example:

$$x(t) = e^{-|t|} \qquad -\infty < t < \infty$$

## Discrete Time Signal

Only defined at specific values of time.

Example:

$$x(t_n) = e^{-|t_n|} \qquad n = 0, \pm 1, \pm 2, \pm 3, \ldots$$

## Continuous Valued Signal

Takes on all possible values on a finite or an infinite range.

Example:

$$x(n) = e^{-|n|} \qquad n = 0, \pm 1, \pm 2, \pm 3$$

## Discrete Valued Signal

Takes on values on a finite or an infinite range.

Example:

## Digital Signal

Discrete Time Signal + Discrete Valued Signal

## Analog Signal

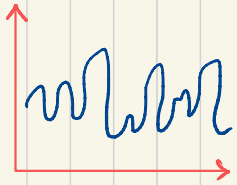
Continuous Time Signal + Continuous Valued Signal

Sampling : Continuous Time → Discrete Time

Quantization : Continuous Valued → Discrete Valued

# Analog to Digital

Analog signal

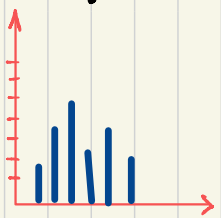


$x_a(t)$

Sampler

(high or enough sampling rate)

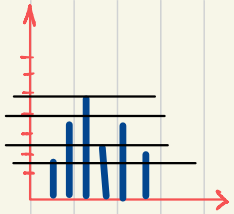
Discrete Time Signal



$x(n)$

Quantizer

Quantized Signal



$x_q(n)$

coder

binarize

## Steps

1. Input → analog signal
2. **Sampling**

   Taking values after every t unit time.
3. **Quantization**

   Continuous Valued Signal → Discrete Valued Signal by rounding or truncating. (Approximation).

## Deterministic Signal

Explicit mathematical expression, predictable, known.

Ex: Sign curve

## Random Signal

Lack of relationship, signals exhibit unpredictable behavior.

Ex: Noise generator, seismic signal.

## Continuous Time Sinusoidal Signal

Simple Harmonic Oscillation

$$x_a(t) = A\cos(\Omega t + \theta) \qquad -\infty < t < \infty$$

$$x_a(t) = A\cos(2\pi F t + \theta) \qquad [\Omega = 2\pi F]$$

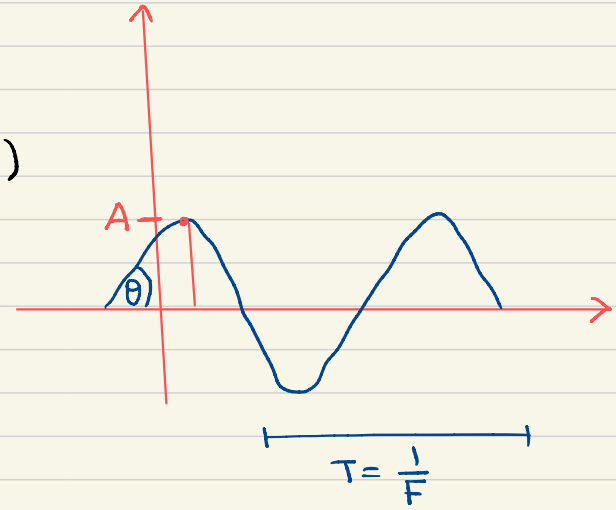
Analog Signal, $x_a(t)$

Amplitude, A

Angular Frequency, $\Omega$ (rad/s)

Frequency, F ($H_z$) (cycles/s)

Phase, $\theta$ (rad)




# A signal is periodic if $x_a(t + T_p) = x_a(t)$

Proof:

$$x_a(t+T_p) = A\cos(2\pi F(t+T_p) + \theta)$$

$$= A\cos(2\pi F T_p + 2\pi F t + \theta)$$

$$= A\cos\{2\pi + (2\pi F t + \theta)\}$$

$$= A\cos(2\pi F t + \theta)$$

$$= x_a(t)$$

Fundamental/Full period $T_p = 1/F$.

# Distinct

N → Period

## Discrete Time Sinusoidal Signal

$$x(n) = A\cos(\omega n + \theta) \qquad n = 0, \pm 1, \pm 2, \ldots$$

$$x(n) = A\cos(2\pi f n + \theta)$$

Digital Signal, $x(n)$

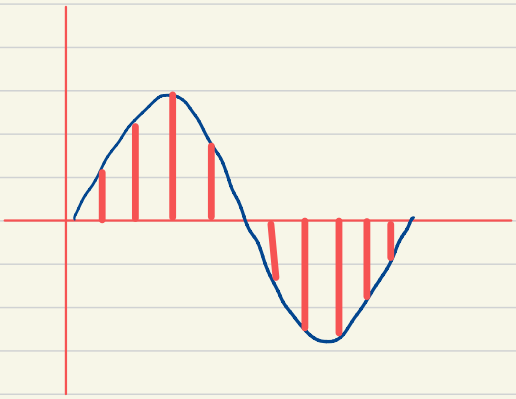
Amplitude, A

Sample Number, n

Angular Frequency, $\omega$ (rad/s)

Frequency, f (cycles/s)

Phase, $\theta$ (rad)



# Periodic only if f is a rational number

$x(n+N) = x(n)$ for all n.

Proof:

$$x(n+N) = A\cos(2\pi f(n+N) + \theta)$$

$$= A\cos(2\pi f n + 2\pi f N + \theta)$$

$$= A\cos\{2\pi k + (2\pi f n + \theta)\}$$

$$= A\cos(2\pi f n + \theta)$$

rational, $f = \frac{k}{N}$

# Discrete time signals whose frequencies are separated by an integer multiple of $2\pi$ are identical.

# Max frequency at $\omega = \pi, -\pi$, $f = \frac{1}{2}, -\frac{1}{2}$.

#

$\omega_0 = 0$

# Recap

## Continuous Time Signal

$$x_a(t) = A\cos(2\pi Ft + \theta) \qquad -\infty < \text{time}, t < \infty$$

## Discrete Time Signal

$$x(n) = A\cos(2\pi fn + \theta) \qquad -\infty < \text{sample}, n < \infty$$

$$n = 0, \pm 1, \pm 2, \pm 3, \ldots$$

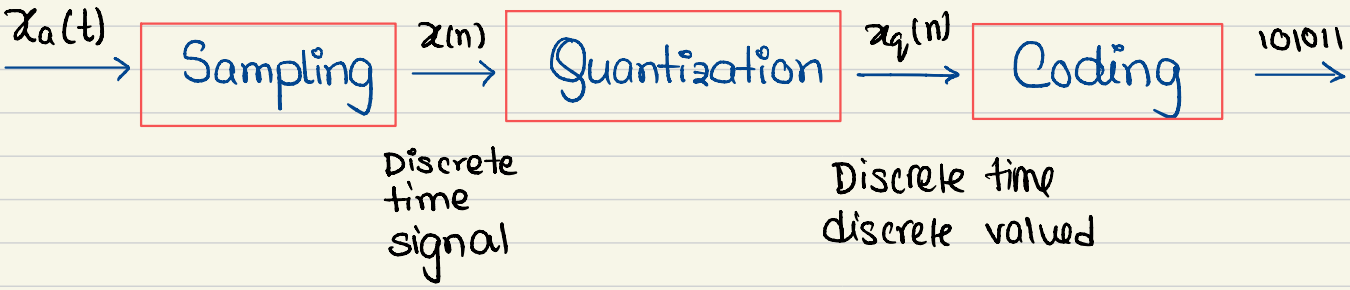
| Continuous-time signals | | Discrete-time signals | |
|---|---|---|---|
| $\Omega = 2\pi F$ | | $\omega = 2\pi f$ | |
| $\frac{\text{radians}}{\text{sec}}$ | Hz | $\frac{\text{radians}}{\text{sample}}$ | $\frac{\text{cycles}}{\text{sample}}$ |
| | $\omega = \Omega T, f = F/F_s$ | $-\pi \le \omega \le \pi$ | $-\frac{1}{2} \le f \le \frac{1}{2}$ |
| | $\Omega = \omega/T, F = f \cdot F_s$ | | |
| $-\infty < \Omega < \infty$ | | $-\pi/T \le \Omega \le \pi/T$ | |
| $-\infty < F < \infty$ | | $-F_s/2 \le F \le F_s/2$ | |

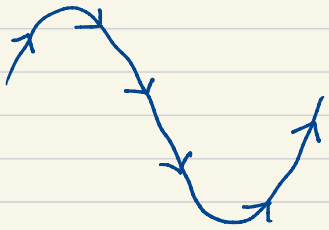
$$F_{max} = \frac{F_s}{2} = \frac{1}{2T}$$

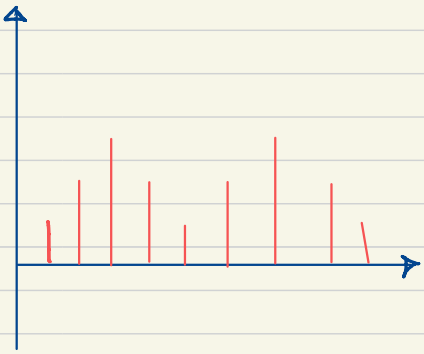
$$\Omega_{max} = \pi F_s = \frac{\pi}{T}$$

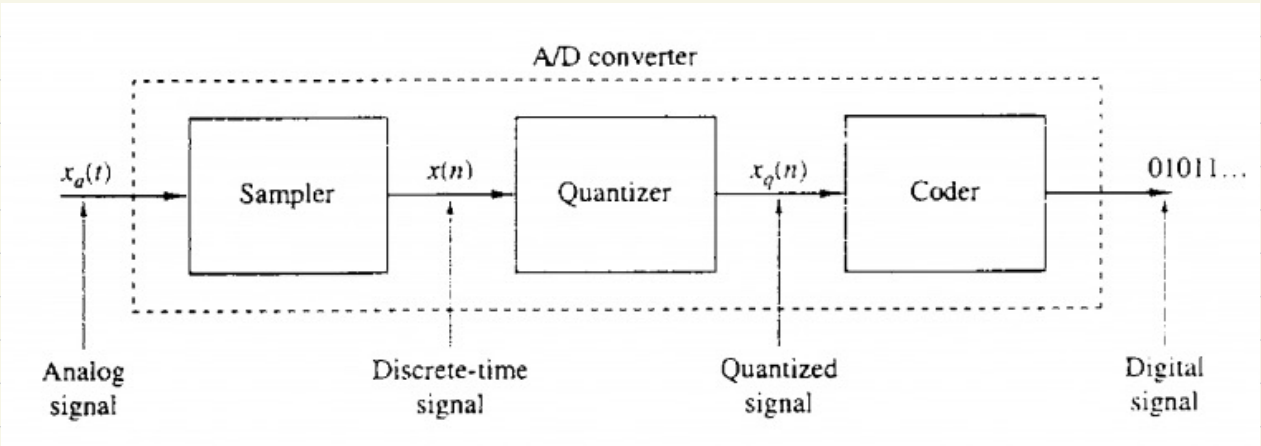
# Analog to Digital Conversion

1. Sampling
2. Quantization (non-invertible)
3. Coding







$$\text{Folding Freq} = \frac{F_s}{2} = F$$

## Sampling

$$x(n) = x_a(nt) \qquad -\infty < n < \infty$$

Sampling period / interval, $T$

Sampling rate, $F_s = 1/T$ (samples/sec)

$$t = nT = \frac{n}{F_s}$$

$$x(n) = x_a(nT)$$

$$= A\cos(2\pi F n T + \theta) \qquad nT = \frac{n}{F_s}$$

$$= A\cos\left(2\pi F \cdot \frac{n}{F_s} + \theta\right)$$

$$\Rightarrow A\cos(2\pi f n + \theta) = A\cos\left(2\pi \frac{F}{F_s} n + \theta\right)$$

$$\therefore \quad \boxed{f = \frac{F}{F_s}} \qquad \omega = \Omega T$$

\# $$-\frac{1}{2} < f < \frac{1}{2} \qquad \left[-\pi < \omega < \pi\right]$$

$$\Rightarrow -\frac{1}{2} < \frac{F}{F_s} < \frac{1}{2}$$

$$\Rightarrow -\frac{F_s}{2} < F < \frac{F_s}{2}$$

$$\Rightarrow -\frac{1}{2T} = -\frac{F_s}{2} < F < \frac{F_s}{2} = \frac{1}{2T}$$

$2F \leq F_s$ otherwise, cant be reconstructed

cont discrete

# Example 1.4.1

The implications of these frequency relations can be fully appreciated by considering the two analog sinusoidal signals

$$x_1(t) = \cos 2\pi(10)t$$
$$x_2(t) = \cos 2\pi(50)t \qquad (1.4.12)$$

$$x_1(t) = \cos 2\pi\left(\frac{1}{4}\right)n = \cos\frac{\pi}{2}n$$

$$x_2(t) = \cos 2\pi\left(\frac{5}{4}\right)n = \cos\frac{5\pi}{2}n$$
$$= \cos\frac{\pi}{2}n$$

# Alias frequency

10, 50, 90, 130, ...

Analog of this freq will produce same discrete signal at 40 Hz sampling rate.

Pg 27-28

An infinite number of continuous-time sinusoid is represented by sampling the same discrete time signal.

Folding frequency $F_s/2$, $\omega = \pi$

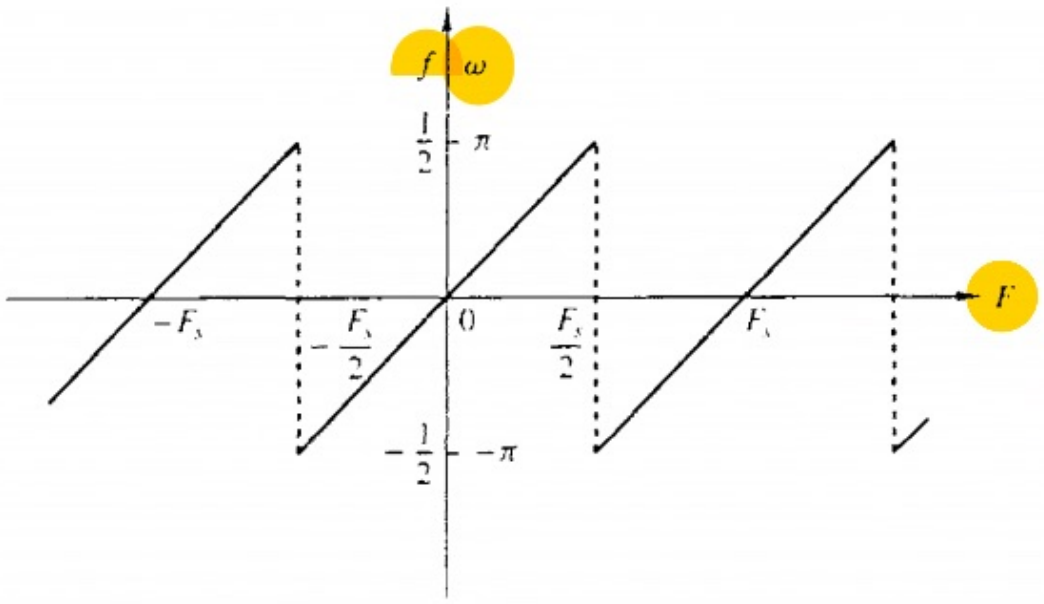



**Figure 1.17** Relationship between the continuous-time and discrete-time frequency variables in the case of periodic sampling.

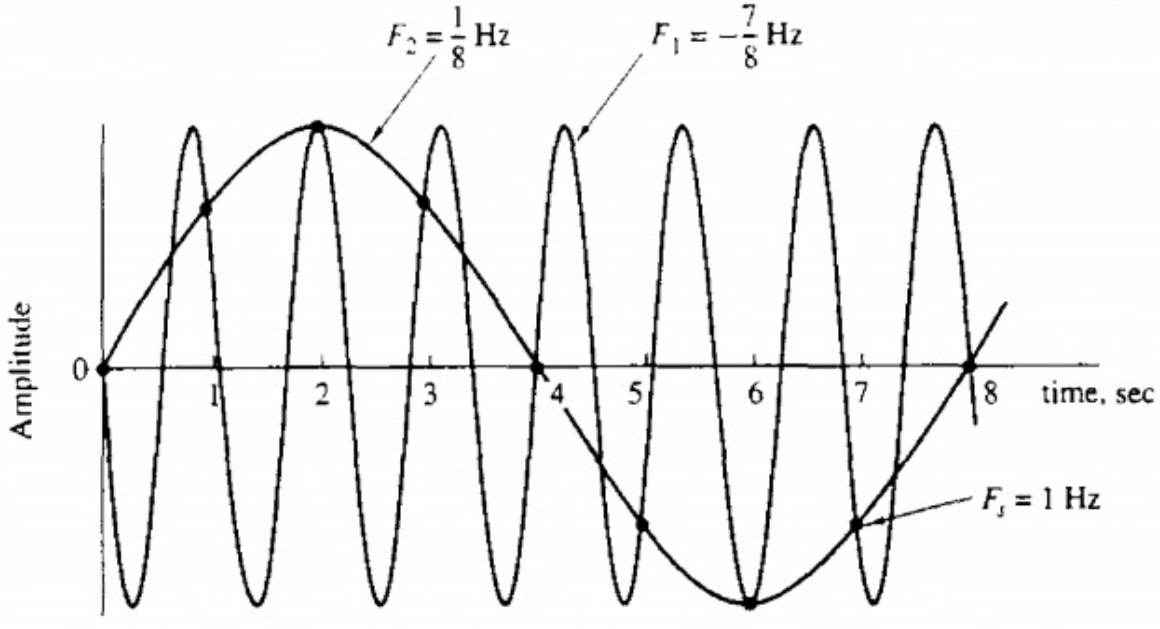



**Figure 1.18** Illustration of aliasing.

$A\cos(2\pi f n)$ $\quad F = 50$

**Example 1.4.2**

Consider the analog signal

$$x_a(t) = 3\cos 100\pi t$$

**(a)** Determine the minimum sampling rate required to avoid aliasing.

**(b)** Suppose that the signal is sampled at the rate $F_s = 200$ Hz. What is the discrete-time signal obtained after sampling?

**(c)** Suppose that the signal is sampled at the rate $F_s = 75$ Hz. What is the discrete-time signal obtained after sampling?

**(d)** What is the frequency $0 < F < F_s/2$ of a sinusoid that yields samples identical to those obtained in part (c)?

a) $F_s \geq 2F \quad \therefore F \geq 100$

b) $\frac{F}{F_s} = \frac{1}{4}$

$$\therefore x(n) = 3\cos\left(2\pi \tfrac{1}{4} n\right)$$

$$= 3\cos\left(\tfrac{\pi}{2}\right)n$$

c) $\frac{F}{F_s} = \frac{2}{3}$

$$\therefore x(n) = 3\cos\left(2\pi \cdot \tfrac{2}{3} n\right)$$

$$= 3\cos\left(\tfrac{4\pi}{3} n\right)$$

$$= 3\cos\left(\tfrac{2\pi}{3} n\right)$$

d) $F_s = 75\ H_2$ , $f = \frac{1}{3}$ $\qquad f = \frac{F}{F_s}$ $\qquad \therefore F = 25\ H_2$

$\therefore F = 50\ H_2$ is an alias of $F = 25\ H_2$ at $F_s = 75$.

## Sampling Theorem

$F_s \geq 2 F_{max}$

**Example 1.4.3**

Consider the analog signal

$$x_a(t) = 3\cos 50\pi t + 10\sin 300\pi t - \cos 100\pi t$$

What is the Nyquist rate for this signal?

## Solution

$F_{max} = 150$

$\therefore F_s \geq 300$ Hz.

**Example 1.4.4**

Consider the analog signal

$$x_a(t) = 3\cos 2000\pi t + 5\sin 6000\pi t + 10\cos 12.000\pi t$$

**(a)** What is the Nyquist rate for this signal?

**(b)** Assume now that we sample this signal using a sampling rate $F_s = 5000$ samples/s. What is the discrete-time signal obtained after sampling?

**(c)** What is the analog signal $y_a(t)$ we can reconstruct from the samples if we use ideal interpolation?

Solution :

a) $F_{max} = 6000 \text{ Hz}$

$\therefore F_S \geq 12000 \text{ Hz}$

b) $f = \frac{F}{F_S} = \frac{F}{5000}$

$\therefore x(n) = 3\cos 2\pi\left(\frac{n}{5}\right) + 5\sin 2\pi\left(\frac{3n}{5}\right) + 10\cos 2\pi\left(\frac{6n}{5}\right)$

$= 13\cos 2\pi\left(\frac{n}{5}\right) - 5\sin 2\pi\left(\frac{2\pi}{5}\right)$

## Quantization

Discrete Time Continuous Signal → Digital signal

### Issues

1. Loss of signal
2. Alias ambiguity

10 Hz
40 Hz
> same

**Quantization Step (Resolution)**, $\Delta = \frac{x_{max} - x_{min}}{L - 1}$

L = # no. of levels
↑ accuracy
↓ info lost

## Quantization Error

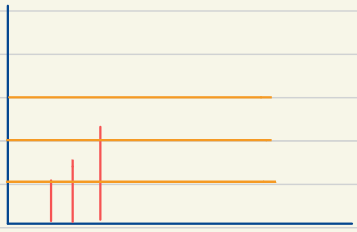
Less if resolution is less.

$$SQNR = \frac{3}{2} 2^{2b}$$

$b \rightarrow$ # of bits

$$SQNR(dB) = 10 \log_{10} SQNR$$

Discrete Timed  Discrete Valued

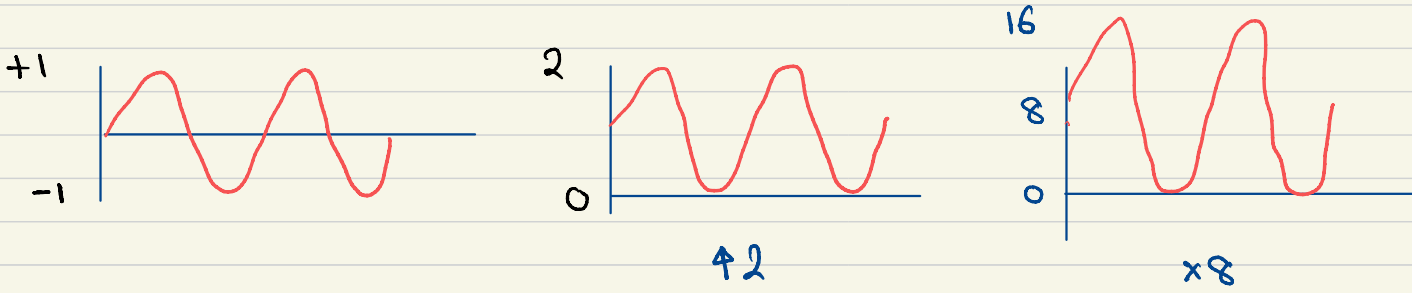


Truncate and Round / → levels

Encode – $\log_n$ (levels)

$\log_2 10 \rightarrow$ 4 bits (approx.)

$(-1, 1) \rightarrow (0, 16) \longrightarrow$ Easy to binarize




Normalize

Encoded → (-1, 1)

$$x_n = \frac{x - a}{b - a} \times (d - c) + c$$

$x - a$: take to 0

$b - a$: scale

$(-1, 1) \longrightarrow (0, 16)$

$a : b$ → $c : d$

[Not in book]

# PROAKIS, CH-2

## Discrete Time Signal

Function of an independent variable, an integer.

### 1. Functional representation

$$x(n) = \begin{cases} 1, & n = 1, 3 \\ 4, & n = 2 \\ 0 & o/w \end{cases}$$

### 2. Tabular representation

| $n$ | -2 | -1 | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|---|---|
| $x(n)$ | 0 | 0 | 0 | 1 | 4 | 1 | 0 |

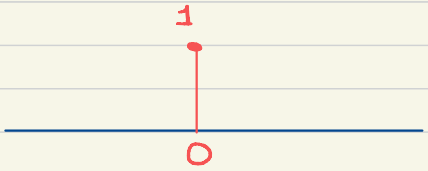
### 3. Sequence representation

$$x(n) = \{ \ldots, 0, \underset{\uparrow}{1}, 4, 1, 0, \ldots \}$$

# Elementary Discrete Time Signal

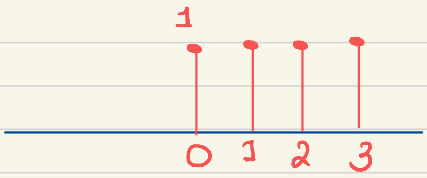
## 1. Unit Sample:

$$\delta(n) = \begin{cases} 1, & n=1 \\ 0 & \text{o/w} \end{cases}$$



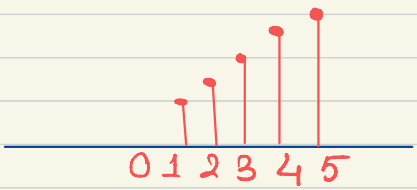

## 2. Unit Step:

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0 & \text{o/w} \end{cases}$$




## 3. Unit Ramp:

$$u_r(n) = \begin{cases} n, & n \geq 0 \\ 0 & \text{o/w} \end{cases}$$



Energy signal

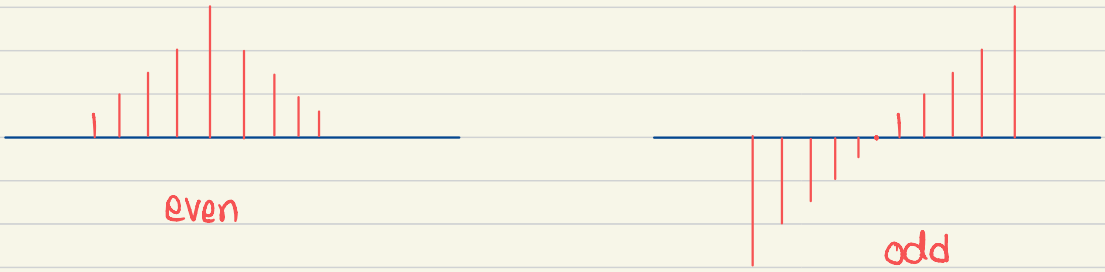
$$E \equiv \sum_{n=-\infty}^{\infty} |x(n)|^2$$

Periodic Signal → Rational $\qquad f_0 = \frac{K}{N}$

Aperiodic Signal → Irrational

Symmetric/Even Signal $\qquad x(n) = x(-n)$

Antisymmetric/Odd Signal $\qquad x(n) = -x(-n)$




"Any discrete time signal can be shown as summation of one odd and one even signal."

– Even Odd Decomposition

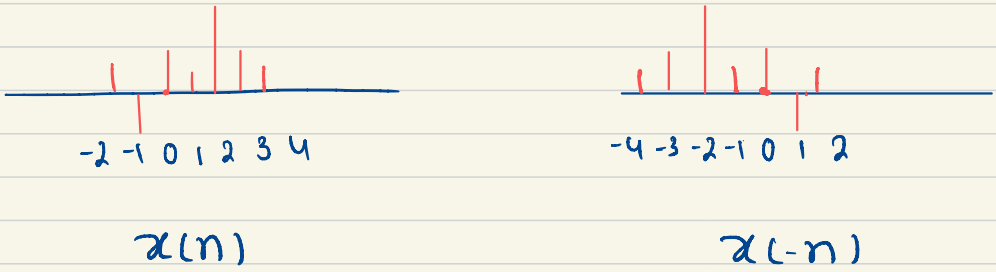
$$x(n) = x_e(n) + x_o(n)$$

$$x_e(n) = \frac{1}{2}\left[x(n) + x(-n)\right]$$

$$x_o(n) = \frac{1}{2}\left[x(n) - x(-n)\right]$$

## Folding

Reflected about $n=0$ $(y=0)$ $\longrightarrow$ $x(n)$



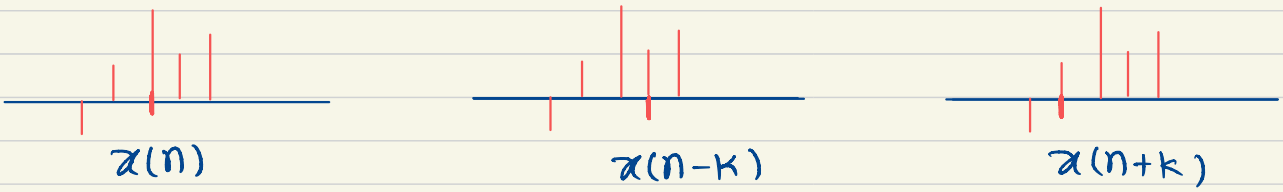

$x(n)$ $\qquad$ $x(-n)$

## Shifting

Sliding left/right

$x(n-k)$ $\longrightarrow$ shift right $(-)$

$x(n+k)$ $\longrightarrow$ shift left $(+)$



$x(n)$ $\qquad$ $x(n-k)$ $\qquad$ $x(n+k)$

Folding : Virtual Delay

Shifting : Time Delay

} Order Matters.

# $x(-n+2)$

$x(-(n-2))$

mirror, right shift

## Down Sampling

Time scaling

$$y(n) = x(2n)$$

$2^{th}$ of original → $1^{th}$ of down sampled.

* Sample to sample basis

Sampling Rate $\frac{1}{T}$ to $\frac{1}{nT}$

## # Discrete Time System (Pg 55)

$$y(n) = \frac{1}{3}[x(n+1) + x(n) + x(n-1)]$$

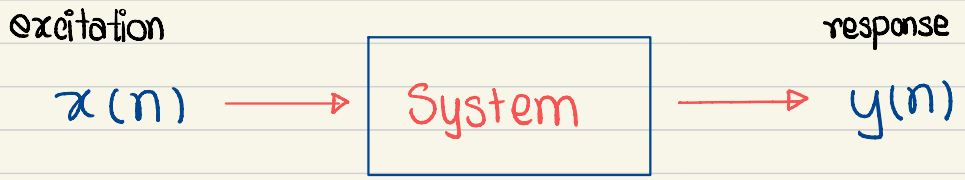
Moving Avg Filter

↳ smoother difference
decrease sharpness

Accumulator → Sum

Math above 2.2.2

# Discrete Time Systems




a) Identity, $y(n) = x(n)$

b) Delay, $y(n) = x(n-1)$

c) Advance, $y(n) = x(n+1)$

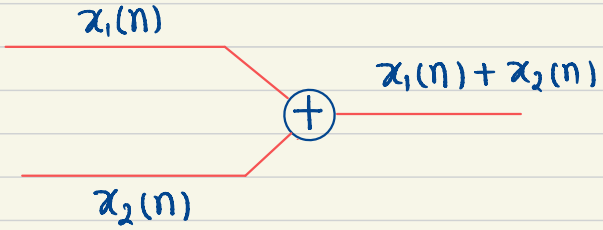
d) Running Average, $y(n) = \frac{1}{3}[x(n-1) + x(n) + x(n+1)]$

e) Maximum, $y(n) = \max[x(n-1), x(n), x(n+1)]$

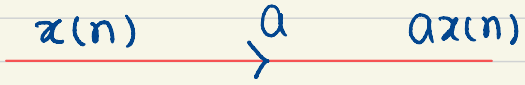
f) Accumulator, $y(n) = \sum_{K=-\infty}^{\infty} x(k)$

# Block Representation

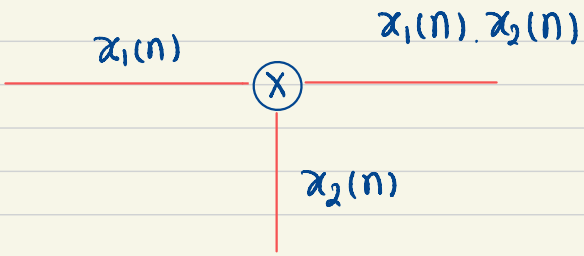
## 1. Adder



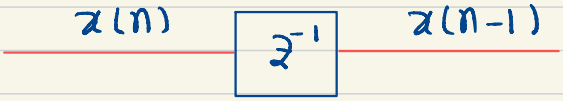

## 2. Constant Multiplier




## 3. Signal Multiplier



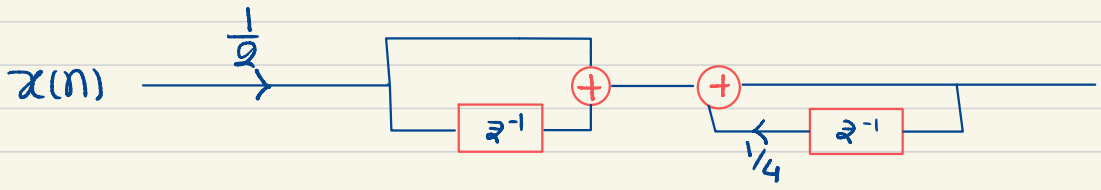

## 4. Unit Delay




## 5. Unit Advance

## Example 2.2.3

$$y(n) = \frac{1}{4} y(n-1) + \frac{1}{2} x(n) + \frac{1}{2} x(n-1)$$




Superposition → Homogeneity + Addetivity [Linear]

Linear, $T[a_1 x_1(n) + a_2 x_2(n)] = a_1 T[x_1(n)] + a_2 T[x_2(n)]$

Causal vs Non-causal

↓

Present & Past

# Resolution of a Discrete Time Signal into impulses

Impulse = $\{0, \underset{\uparrow}{0}, 1, 0\}$ Spike

Unit Impulse = $\{0, 0, \underset{\uparrow}{1}, 0, 0\}$

0th sample = 1

"Every signal is a summation of bunch of shifted & scaled Unit Impulses"

$x(n) = \{2, \underset{\uparrow}{4}, 0, 3\}$

$= \{2, \underset{\uparrow}{0}, 0, 0\} = 2\,\delta(n+1)$

$= \{0, \underset{\uparrow}{4}, 0, 0\} = 4\,\delta(n)$

$= \{0, \underset{\uparrow}{0}, 0, 0\} = 0\,\delta(n-1)$

$= \{0, 0, \underset{\uparrow}{0}, 3\} = 3\,\delta(n-2)$

} Decomposed

$\{2, 4, 0, 3\}$

$m\,\delta(n+k)$

m → scale

k → shift

$$E_K = \sum_{-N}^{N} |x(n)|^2$$

$$P = \frac{1}{2N+1} E_N$$